

Lecture 10: Applications and Policy Design

Bringing It All Together — When, Where, and How Behavioural Economics Works

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The Central Question of Behavioural Policy

If people are systematically biased in predictable ways, what should policymakers do?

1. **Inform:** Tell people about their biases and let them correct themselves.
2. **Incentivise:** Change prices and subsidies to align incentives.
3. **Nudge:** Design the choice environment to exploit known biases for good.
4. **Mandate:** Remove the choice entirely and impose the right outcome.

Today's Lecture

We apply the full toolbox of behavioural economics to four domains: **health, environment, finance, and tax compliance**. We also examine the **ethics and limits** of behavioural policy — and assess the field's future.

The Policy Design Matrix

Tool	Mechanism	When Best	Canadian Example
Information	Disclosure; education	When bias = lack of info	Calorie labels, FCAC
Price/Tax	Change financial incentive	High price elasticity	Carbon levy, cig. taxes
Nudge	Defaults, framing, salience	Present bias / inertia	RRSP auto-enrol, organ donation
Subsidy	Reduce cost	Financial barrier binding	Canada Learning Bond
Mandate	Legal requirement	Large externalities	Helmet laws, car seats

Key Insight

The tools complement each other. The best policy often *combines* approaches — a carbon levy (price) + awareness campaign (information) + green energy defaults (nudge).

Information campaigns alone are weak when the problem is present bias. 70% of smokers *know* smoking causes cancer.

What Works (RCT Evidence)

- **Commitment contracts:** Gym contracts +20% attendance (Acland & Levy, 2015).
- **Implementation intentions:** “When/where will you get your shot?” $\Rightarrow$ +13% uptake.
- **Social norms:** “Most patients take medication as prescribed” $\Rightarrow$ +8%.

Canada-Specific Successes

- **BC QuitNow:** 23% quit rate vs 5% unassisted.
- **Nova Scotia opt-out (2021):** registration rising.
- **BC CDC COVID messaging:** loss-framed prevention + gain-framed testing $\Rightarrow$ 2nd lowest mortality in Canada (2020).

Vaccination: Why People Don't Vaccinate

Five Biases Working Against Vaccination

1. **Present bias:** Benefit (disease prevention) is future and uncertain; cost (going to clinic) is now and certain.
2. **Optimism bias:** “I won’t get the flu.” Overconfidence in own immunity — a form of above-average effect.
3. **Loss aversion:** Fear of side effects is *salient and immediate*; disease prevention is *abstract and future*.
4. **Availability heuristic:** If nobody in your social network got the flu last year, perceived risk is low.
5. **Descriptive social norms:** If peers don’t vaccinate, the descriptive norm actively discourages vaccination.

Key Insight

The problem is **not lack of information** — 70%+ of Canadians know vaccines work. It is a collection of biases pulling in the opposite direction.

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Milkman et al. (2021) — 47,000-Person RCT at a US Health System

- **Reminder only:** +4% uptake vs control.
- **Reminder + Implementation Intention** (“When/where will you go?”): +13% uptake.
- **Ownership frame** (“Your flu shot is waiting for you”): +11% uptake.
- **Reminder + pre-booked appointment:** +15% uptake.
- **Cost of all interventions:** \$0. (Only wording on a text message.)

Canadian Applications

- **COVID-19 messaging (BC CDC, 2020):** Loss-framed prevention messages + gain-framed testing messages $\Rightarrow$ 2nd lowest mortality rate in Canada.
- **BC Pharmacare flu programme:** Text reminder with “your shot is available at [pharmacy name]” $\Rightarrow$ 18% uptake increase (2022 pilot).

Standard tools face limits: carbon taxes face loss-aversion resistance; subsidies are costly; cap-and-trade is complex.

Behavioural Tools

- **Social norms:** Opower reports $\Rightarrow$ 2% reduction = 1 power plant per 6M users.
- **Green default:** 69% opt-out vs 7% opt-in.
- **Commitment:** BC Hydro Team Power Smart $\Rightarrow$ 10%+ reduction.
- **Salience:** Real-time CO₂ on Nest/Ecobee.

BC Case Study

BC carbon tax (2008): revenue-neutral framing + gain frame for rebates + explicit per-household calculation $\Rightarrow$ became **broadly popular**. Lesson: *frame determines political feasibility*.

Food Choices and Behavioural Policy

Three Approaches: Meat Consumption Reduction

Approach	Intervention	Effect
Information	Pamphlet on environmental cost of meat	2% reduction
Price	20% meat tax	7% reduction
Nudge (default)	Plant-based option as canteen default	15% reduction
Nudge (salience)	Green leaf label on plant-based items	5% reduction
Mandate	No meat options at all	100% (backlash risk)

Denmark Case Study

Danish hospitals set plant-based food as the default in staff canteens (2013). Plant-based consumption rose **400%**. Zero complaints. Food waste fell 35%. Canadian hospital food service: similar pilots proposed for BC hospital cafeterias (2023 proposal to BC Ministry of Health).

Canadian regulators: FCAC (fee disclosure, min-payment warnings) · OSFI (mortgage stress testing B-20, 2018 — commitment device) · CIPF (investment dealer disclosure).

Key Principles

1. **Salience:** One-page “Key Facts” for mortgages (2022).
2. **Defaults:** Auto-enrollment over mandates.
3. **Friction reduction:** CRA pre-fill.
4. **Sludge removal:** Easy cancellation.

CRM2 Reforms (2016)

Required disclosure of (1) fees in dollars; (2) performance vs benchmark. Fee awareness soared. Shift to ETFs accelerated: Canadian ETF AUM \$80B (2016) → **\$350B** (2023).

Why People Don't Pay Their Taxes (Behavioural Perspective)

- **Present bias:** Payment is now; enforcement is uncertain and future.
- **Social norms:** If you believe others are evading, social constraint is relaxed.
- **Complexity:** Cognitive overload from complex forms $\Rightarrow$ errors and avoidance.
- **Loss aversion:** A tax payment is experienced as a loss (even though taxes are obligations).

The BIT's HMRC Experiments (UK Tax Authority)

- Adding social norm: +15% payment rate.
- Simplifying the letter: +5% payment rate.
- Adding implementation intention: "Pay online in 3 minutes at hmrc.gov.uk/pay" $\Rightarrow$ +8%.
- Combined: +**28%** vs standard letter. Revenue impact: hundreds of millions of pounds.
- CRA adoption of similar strategies: +\$200M estimated annual revenue at full scale.

The Criticisms

1. **Manipulation:** Exploiting biases vs helping people reason better.
2. **Autonomy:** Infringes self-determination.
3. **Paternalism:** Who decides? Government can be wrong.
4. **Scale:** Small effects; systemic problems need systemic solutions.
5. **Corruption:** Corporations use same techniques against consumers.

Thaler (2018): Nudges are ethical when transparent, opt-outable, evidence-based, and benefit the chooser.

The Defenders' Response

1. Transparency + opt-out = legitimate.
2. Someone always designs the environment.
3. Paternalism can be evaluated and reversed.
4. Small effects $\times$ scale = large aggregate benefit.
5. Regulate corporate nudges separately.

When Do Nudges Fail?

Limits of the Nudge Approach

1. **Structural vs behavioural problems:** If people can't save because wages are too low, a default doesn't help. Nudges work within a given constraint set — they don't change the constraints.
2. **Effect size:** Many nudge effects are statistically significant but economically small (1–3%). Systemic health or environmental challenges require larger interventions.
3. **Persistence:** Some nudge effects fade over time. Auto-enrollment raises participation but not always contribution rates.
4. **Heterogeneity:** One default cannot fit all. Personalised nudges are more effective but harder to implement at scale.
5. **Adaptation:** Sophisticated individuals (often higher-income) learn to ignore or reverse nudges. This can make nudges regressive.

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The Welfare Question

With time-inconsistent preferences, there are **three** selves: **Instant** ($\beta < 1$: wants cake now); **Long-run** ($\beta = 1$: wants health); **Reflective** (may endorse either).

O'Donoghue & Rabin (2003)

Paternalistic correction of present bias is welfare-improving if the long-run preference is the “true” preference — not always clear for addiction and lifestyle choices.

The Libertarian Critique (Sugden, 2017)

Even if people are biased, it doesn't follow that government knows better. Governments have their own biases and political incentives. Behavioural policy could become a tool for political manipulation.

Final Group Challenge — 8 Minutes

Canada's federal government asks you to design a behavioural policy intervention for one problem:

- **Group A:** Increase RRSP contributions among 25–40 year olds.
- **Group B:** Reduce single-use plastic in university cafeterias.
- **Group C:** Increase workplace vaccination rates to 90%+.
- **Group D:** Reduce payday loan usage in low-income communities.

Your intervention must: (1) identify the key bias, (2) propose one nudge and one complementary intervention, (3) describe your RCT evaluation design.

The Future of Behavioural Economics

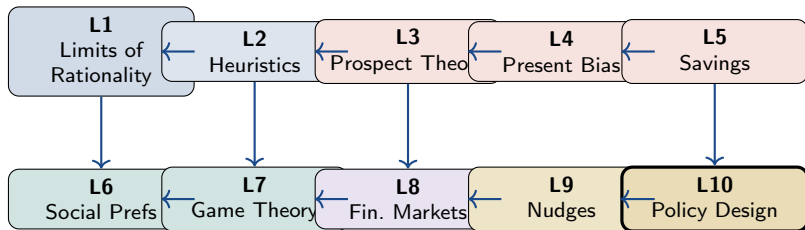
Emerging Frontiers

1. **Personalised nudges:** Machine learning to match nudge to individual bias profile. Amazon, Spotify already do this commercially.
2. **Neuroeconomics:** fMRI identifies neural mechanisms; predicts who responds to which nudge.
3. **Digital nudges:** Notifications, app design, smart defaults. The smartphone is the greatest nudge delivery platform ever invented.
4. **International development:** JPAL (Banerjee, Duflo) applies behavioural tools in low-income countries. Nobel Prize 2019.
5. **Climate change:** Behavioural public economics integrated with climate policy. Paris Agreement pledges: present-biased governments.

The Reproducibility Debate

Camerer et al. (2018): 61% of 18 replication attempts of lab results succeeded. Loss aversion, present bias, anchoring: **high replication**. Some social priming effects: **failed**. Lesson: policy should rely on highly replicated findings.

The Course in One Diagram



The Core Message

"People are not irrational. They are intentionally rational — but cognitively constrained. Understanding those constraints is how we build a better world."

What You Should Be Able to Do

- **Identify** which bias is operating in a given real-world situation.
- **Predict** how a present-biased, loss-averse, overconfident agent would behave.
- **Design** a nudge that addresses a specific behavioural failure — with justification.
- **Evaluate** a policy using cost-benefit analysis that accounts for behavioural responses.
- **Critique** nudge-based policy using autonomy, effectiveness, and distributive arguments.
- **Apply** Canadian data and context to global behavioural economic findings.

One Last Thought

Behavioural economics does not replace standard economics. It *enriches* it. The standard model tells you what perfectly rational agents would do — the ideal benchmark. Behavioural economics tells you what real people actually do — and how to close the gap. Together, they are the most powerful toolkit in modern social science.

Key Takeaways — Lecture 10 & The Course

- ✓ **Policy toolkit:** information, price, nudge, subsidy, mandate — different tools for different biases.
- ✓ **Health policy:** implementation intentions, commitment contracts, and loss framing outperform information campaigns alone.
- ✓ **Environmental policy:** green energy defaults, social norms, and carbon levy framing.
- ✓ **Financial regulation:** salience, friction reduction, sludge removal, CRM2 disclosure.
- ✓ **Tax compliance:** social norms + simplification = hundreds of millions in additional revenue.
- ✓ **Ethics:** nudges are legitimate when transparent, easy to opt out, evidence-based, and benefit the chooser.
- ✓ **Limits:** nudges don't fix structural problems; effects can be small; personalisation needed.

“The best way to make the world a better place is not to tell people what to do — but to understand why they do what they do.”

— Behavioural Economics, ECON 230, UNBC, Winter 2027

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